

UNDERSTANDING VISUAL BEHAVIOUR IN STROKE PATIENTS USING TOPOGRAPHIC NEURAL-NETWORK MODELS

A THESIS

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CERTIFICATE

This is to certify that the thesis titled "Understanding visual behaviour in stroke patients using Topographic Neural-Network models" is the outcome of original work carried out by Aditi Arun for the degree of Bachelor of Science (Research) with a Major in Biology at the Indian Institute of Science, Bangalore, India.



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DECLARATION

The work reported in this thesis is original and was carried out by me during my tenure as an undergraduate student at the Indian Institute of Science, Bangalore. This thesis has not formed the basis for the award of any degree, diploma, associateship, membership, or similar title of any university or institution.

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Chapter 1

Introduction

Stroke is a major cause of neurological impairment and can lead to a wide range of deficits, including significant changes in vision. Visual impairments are common in stroke survivors and can significantly affect independence and quality of life. These deficits vary depending on the location and extent of the brain injury and may include visual field loss (such as hemianopia, where half the visual field is lost), visual neglect (inattention to one side of space), and difficulties with depth perception or visual acuity. Some patients may also experience impaired eye movements, resulting in double vision or difficulty tracking objects.

When a stroke involves the primary visual cortex, located in the occipital lobe, the resulting visual deficits are often more specific and predictable. This region, also known as V1, is the brain's initial cortical area for processing visual information received from the retina via the thalamus. Damage to V1 commonly causes hemianopia or quadrantanopia, where patients lose part of their visual field on the side opposite the brain lesion. In more severe cases, the damage may lead to cortical blindness, where the eyes and optic nerves remain intact, but the brain can no longer interpret visual input.

Despite considerable research into motor and cognitive deficits resulting from stroke, the literature examining visual impairments remains comparatively limited, particularly regarding computational modelling efforts. A robust and biologically plausible computational

framework capable of accurately simulating post-stroke visual behaviours would enhance our understanding of stroke pathology and guide clinical interventions aimed at rehabilitation and recovery.

In recent years, topographic deep artificial neural networks (TDANNs) have emerged as powerful models for investigating the organization and function of the primate visual cortex [2–4]. TDANNs leverage convolutional neural networks (CNNs) [10] inspired by biological neural networks and have demonstrated remarkable success in capturing key functional properties and spatial organization of visual processing areas. Importantly, these models provide a means to simulate cortical lesions and evaluate resulting behavioural impairments, bridging computational neuroscience and clinical neurology.

1.1. Hypothesis

This thesis hypothesizes that a TDANN, structurally and functionally aligned with the primate visual cortex, can accurately simulate the behavioural deficits observed in human stroke patients following lesions in the primary visual cortex. Specifically, I propose that localized perturbations within the V1 layer of a TDANN will yield measurable, predictable impairments in visual perception tasks, mirroring those observed clinically.

1.2. Objectives

The overarching goal of this thesis is to validate the use of TDANN models to simulate and understand visual impairments resulting from stroke. Towards this goal, I consider the following objectives:

- To identify suitable experimental studies investigating visual deficits following strokes, and curate a comprehensive dataset from the Fahrenthold (2021) study [1], involving behavioural data from Humphrey Visual Field (HVF) tests and lesion size estimations,

to ensure precise and clinically relevant modelling.

- To replicate visual behavioural deficits observed in stroke patients by inducing spatially localized, stroke-like lesions in a TDANN that models the human visual cortex.
- To qualitatively and quantitatively assess the TDANN model’s ability to simulate post-stroke visual behaviours.
- To identify limitations of current TDANN models and provide clear directions for future improvements

1.3. Thesis outline

The rest of this thesis is structured as follows:

- Chapter 2 reviews relevant literature on stroke-induced visual impairments and existing computational models of the visual cortex.
- Chapter 3 describes the TDANN architecture, methodologies for simulating cortical lesions, and a detailed description of the HVF test protocol used to measure behavioural responses in stroke patients.
- Chapter 4 presents detailed results from both qualitative and quantitative analyses of the TDANN simulations in comparison with empirical patient data.
- Chapter 5 discusses the findings’ implications, evaluates the current model’s strengths and limitations, and proposes detailed future directions for extending and refining this research approach.

In summary, this thesis systematically evaluates TDANNs as viable tools for modelling visual deficits resulting from stroke-induced cortical damage. By rigorously comparing model behaviour to clinical observations from stroke patients, this research lays the groundwork for

computational models that can potentially inform future studies investigating cortical lesions and visual perception in stroke.

Chapter 2

Literature Review

2.1. Behavioural studies

A review of behavioural studies investigating visual recognition and perception in stroke patients was conducted to model the visual effects of stroke. The cohort of studies in this area was sparse, as most stroke studies investigate motor deficits [5–7], which are often the most common and debilitating outcome of strokes. In total, 14 studies investigating visual deficits in people with stroke-related or lesion-induced brain injuries were found [1, 18, 24, 26–36].

I assessed each study for its potential suitability as a modelling target. The following criteria guided selection:

- The number of human subjects with lesions surveyed and the number of controls.
- The suitability of the task for modelling by the topographic neural network model. Since the model cannot account for time-dependent effects, experiments with tasks with a temporal component were excluded.
- The availability of the raw behavioural data collected from the experiments in the study.
- The inclusion of detailed information on the lesion locations and sizes in the subjects.

Based on the criteria outlined above, the studies by Spang et al. (2021) [24], Fahrenthold (2021) [1], and Robotham et al. (2023) [18] were initially shortlisted for potential modelling. However, despite considerable efforts, we were unable to utilize the software provided by Spang et al. (2021) to recreate the stimulus images used in their behavioural tasks. In the case of Robotham et al. (2023), we contacted the authors to request access to the raw behavioural and lesion data but were ultimately unable to obtain the necessary materials. Consequently, the dataset from Fahrenthold (2021) was selected as the primary focus for our modelling work.

2.2. The Fahrenthold study

The study evaluated 36 patients with unilateral occipital strokes in the primary visual cortex (V1 area) and 15 controls. 28 of the patients were chronic, that is, they had been more than 6 months since their stroke, while 8 were subacute, that is, less than 6 months since their stroke. Among the chronic patients, 5 patients had suffered a hemorrhagic stroke, while the rest of the 31 patients had an ischemic stroke. The image below illustrates the main difference between the two.

Ischemic strokes occur when there is a blockage in blood vessels, leading to oxygen deprivation in a specific region of the brain. This leads to infarction or death of brain tissue, causing a lesion in the area. However, hemorrhagic strokes occur when a blood vessel ruptures, leading to the leakage of blood into the brain. This leads to a hematoma, which is a localized collection of blood. This exerts excess pressure on the surrounding brain tissue, thereby damaging it. Unfortunately, the study did not specify the identities of the 5 patients who had suffered from hemorrhagic stroke. For our research, I have modeled all the lesions with the characteristics of ischemic strokes, the details of which are described in Sec. 3.4.

Damage to the primary visual cortex, post-stroke causes a visual field defect contralateral to the lesion, known as cortically induced blindness (denoted as CB). Additionally, it has been shown that damage to the occipital cortex in humans post-stroke induces a shrinkage in

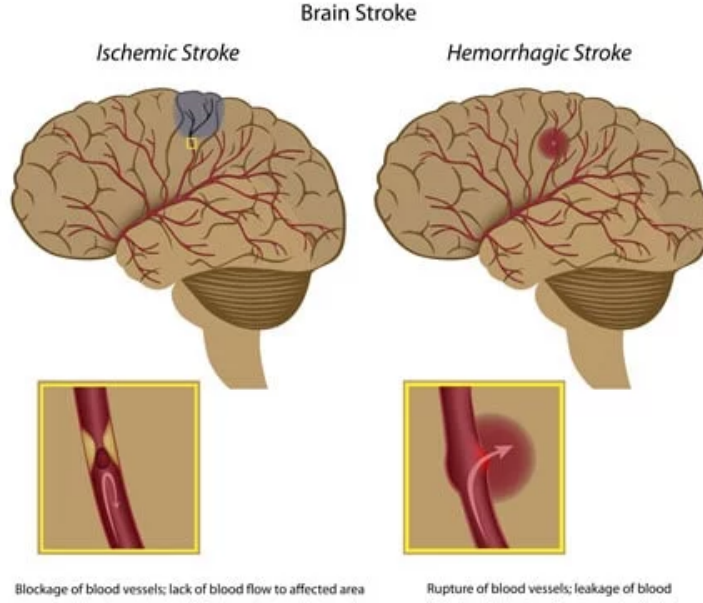


Figure 2.1: Difference between ischemic and hemorrhagic stroke ([Source](#))

the volume of the ipsilateral optic tract. Magnetic resonance imaging was used to measure the optic tract volume, and this was used to compute a laterality index for each subject. I used this information to estimate the lesion sizes for the subjects, further details of which are described in Sec. 3.4. The study investigated the potential for recovery in the subjects when subjected to visual restoration training, and the role optic tract shrinkage plays in modulating this potential. The subjects' visual field recovery was quantified using Humphrey perimetry, measured through the Humphrey Visual Field test (refer to Sec. 3.4). The HVF test was administered both pre- and post-training, but I referred to the Humphrey perimetries of all subjects before any visual restoration training, for our modelling. Lastly, the study did not provide the lesion locations for the subjects, hence multiple lesion locations were tried in order to estimate the best fit to the subject during modelling.

2.3. Simulating the visual cortex

The pioneering experiments by Hubel and Wiesel [8] demonstrated the hierarchical organization of the visual system, showing that simple cells respond selectively to specific orientations

while complex cells integrate these simpler signals to detect more abstract features. These findings inspired early computational models aimed at simulating the primary visual cortex (V1). Notably, Jones and Palmer [9] demonstrated that receptive fields in V1 could be effectively approximated by linear filters combined with nonlinear operations, such as thresholding.

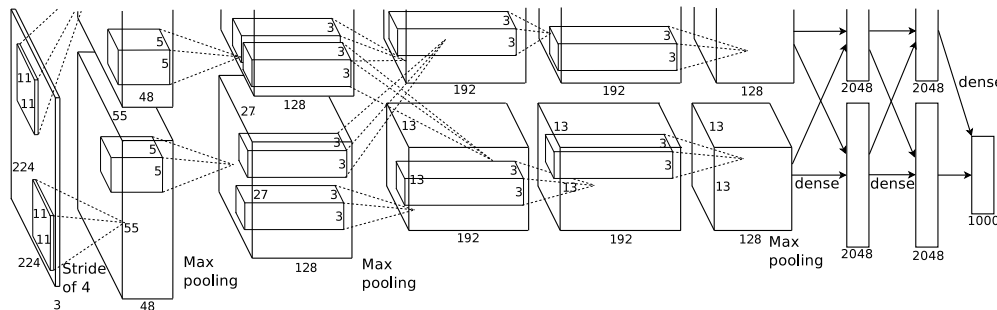


Figure 2.2: The architecture of AlexNet [11] used in ILSVRC 2012.

A significant advancement in modelling the visual cortex came with the resurgence of convolutional neural networks (CNNs) [10, 11], which apply convolutional operations through several hierarchical and nonlinear stages. While early cortical simulation methods typically involved directly fitting network parameters to neural data, more recent developments have revealed the effectiveness of "goal-driven learning." Specifically, training CNNs on ecologically relevant tasks, such as object recognition, significantly enhances the ability of these models to match neuronal responses across cortical areas from V1 to the inferotemporal (IT) cortex (for a comprehensive discussion, see Yamins and DiCarlo [12]). Cadena et al. [13] further demonstrated that deep CNNs trained for object recognition can accurately model V1 neuronal spiking activity in macaques.

Recent work continues to refine these models by incorporating biologically plausible constraints. For instance, the TDANN used in this thesis (described in detail in Sec. 3.3) employs a spatial loss to ensure topographical similarity to the visual cortex and incorporates a self-supervised learning objective, which is more biologically realistic.

Chapter 3

Methods

In this chapter, I describe the topographic deep artificial neural network (TDANN) and how it was used to simulate strokes in the visual cortex. A crucial part of this is gleaning the lesion sizes from the stroke datasets of real patients, described in Sec. 3.4. I conclude with the experimental setup.

3.1. Humphrey Visual Field (HVF) Test

The Humphrey Visual Field (HVF) test is a standardized method widely used to measure visual field sensitivity by systematically assessing the detection thresholds of light stimuli presented at specific retinal locations. In the present study, visual field thresholds of stroke patients were measured using this approach.

HVF Analyzer

The HVF test was conducted using the Humphrey Field Analyzer. It utilizes a bowl-shaped projection surface that subtends 30 or 60 degrees of the visual field, depending on the test protocol, to map the central and peripheral vision of the subject. The analyzer features a concave, dome-shaped projection bowl, typically subtending 30 or 60 degrees of the visual

field, enabling thorough assessment of both central and peripheral vision. During the task, subjects are instructed to fixate on a central target and indicate whether they can see a circular stimulus by pressing a handheld response button. Stimuli are presented in varying locations and luminance intensity, and the subject's responses help the device determine the visual threshold sensitivity at each point.



Figure 3.1: Zeiss Humphrey Visual Field Analyzer ([image source](#)).

In the case of this study, the Humphrey Field Analyzer II-i 750 was used. A standard white stimulus of size III (approximately 0.43 degrees in diameter, or 4 mm² in area) was used, as provided by the Zeiss Humphrey Visual Field Analyzer. This is one of the standard sizes available in the machine and is commonly employed in clinical perimetry for its balance between sensitivity and reliability across various visual field defects. The background luminance was set at 11.3 cd/m², providing a consistent baseline illumination.

All tests were conducted under controlled fixation conditions to ensure the reliability of visual field measurements. Fixation was monitored using real-time gaze tracking, a built-in feature of the Zeiss Humphrey Visual Field Analyzer. Infrared sensors continuously monitored the subject's eye movements and detected deviations from the central fixation point. Additionally, before testing, the visual acuity for each participant was corrected to

20/20 using trial lenses to eliminate refractive error as a confounding factor, ensuring that any deficits measured were due to true field loss rather than blurred vision.

Lastly, an automated algorithm called SITA (Swedish Interactive Thresholding Algorithm) [25] was employed, which dynamically adjusted the stimulus intensity based on prior responses and normative data. This algorithm also quantifies various measures of reliability, including fixation losses, false-positive and false-negative responses, thus providing a quantitative measure of test validity.

Stimulus luminances varied widely, ranging from the brightest at 3183.1 cd/m² (0 dB) to the dimmest at 0.032 cd/m² (50 dB). The luminance values were logarithmically represented in decibels (dB), where 0 dB corresponds to the brightest stimulus detectable by regions of the lowest sensitivity, and 50 dB corresponds to the dimmest stimulus detectable by highly sensitive retinal regions.

Stimulus Presentation

The target stimuli were presented in different locations within a grid-like pattern in the visual field to sample the full space. Two specific testing grids were utilized in the study and are displayed below:

- **24-2 Pattern:** Consisting of 54 locations evenly spaced by 6 degrees, this pattern covers a central visual field area of 42 degrees diameter (see Fig. 3.2).
- **10-2 Pattern:** Comprising 68 locations spaced at 2-degree intervals, this pattern densely samples the central 18 degrees of the visual field (see Fig. 3.3).

Stimuli were presented monocularly, with patients instructed to maintain fixation on a central target to ensure consistent retinal stimulation.

Multiple stimulus luminances were presented for each stimulus location within the grid types to measure the retinal sensitivity at that location. Stimulus luminances varied from

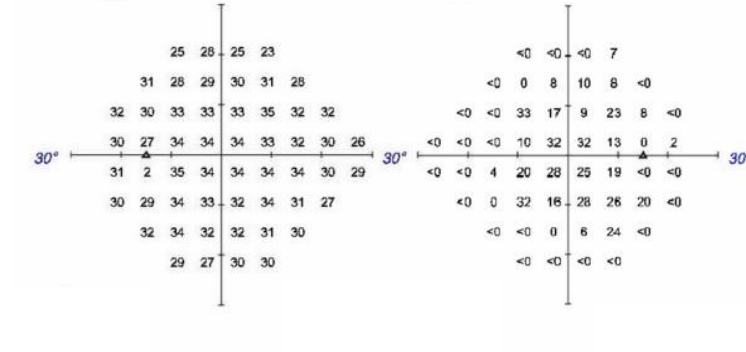


Figure 3.2: The HVF 24-2 grid patterns for the left eye (left) and the right eye (right) respectively. Image adapted from [19].

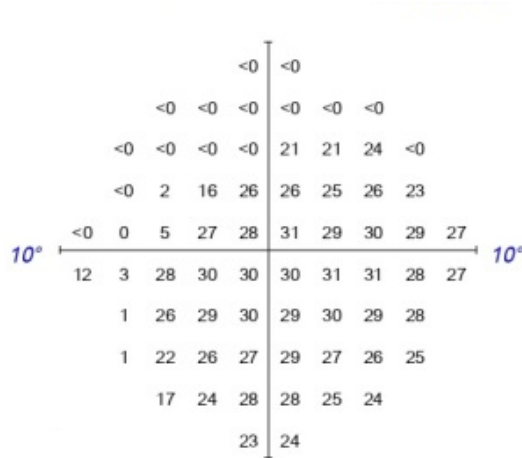


Figure 3.3: The HVF 10-2 grid pattern ([image source](#)).

3183.1 cd/m² to 0.032 cd/m², on a logarithmic scale. Luminance is inversely proportional to the stimulus intensity, which is measured in decibels, and ranges from 0 to 50 dB. Thus, a stimulus intensity of zero decibels (dB) represents the brightest light stimulus, and 50 dB represents the dimmest stimulus. So, a zero decibel stimulus will be visible to a point on the retina with the lowest sensitivity, and vice versa.

Threshold Estimation

Visual field thresholds at each stimulus location were determined using the 4-2-1 staircase method, which involves the following steps:

1. If the initial stimulus at a given location is not perceived, the stimulus intensity is increased in 4 dB increments until detected.
2. Upon detection, the stimulus intensity is reduced in 2 dB decrements until the stimulus is no longer detected.
3. The intensity is then increased in 1 dB increments until the stimulus becomes detectable again.
4. The final intensity level at which the stimulus is reliably detected at least 50%

Thus, a threshold value was calculated for each stimulus location and grid type, and separately for the right and left eyes.

Composite Visual Field Mapping

Visual field threshold maps were constructed by combining data from the 24-2 and 10-2 tests. For each subject, thresholds from the left and right eyes were averaged. Subsequently, measurements from both grid patterns were interleaved, with overlapping locations averaged to provide unified sensitivity estimates. A two-dimensional linear interpolation at a spatial

resolution of 0.1 degrees² was then applied to estimate sensitivity values at untested visual field locations. This resulted in a continuous composite visual field covering approximately 1,616 degrees².

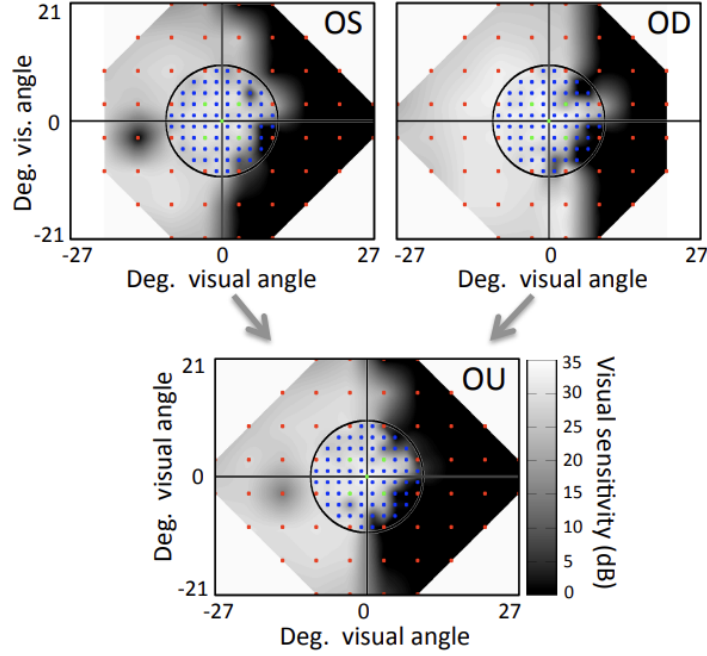


Figure 3.4: Visual field interpolation.

The visual field maps for a few sample subjects in the study are shown in Fig. 3.5.

3.2. Estimating lesion size

As outlined previously, Fahrenthold [1] utilized MRI imaging to measure optic tract volumes and derived a laterality index (LI) defined as:

$$LI = \frac{cOT - iOT}{cOT + iOT} \quad (3.1)$$

Here, cOT and iOT represent the contralesional and ipsilesional optic tract volumes, respectively. The laterality index quantifies the asymmetry between these two sides in a

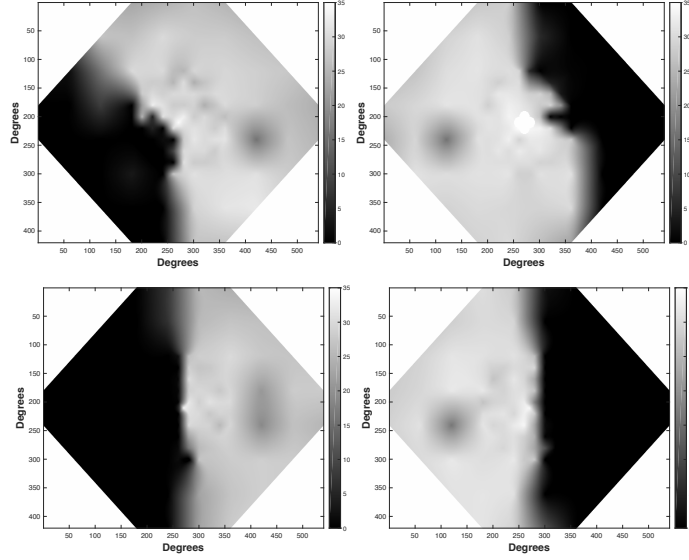


Figure 3.5: Sample visual field maps for 4 subjects in the study.

normalized manner.

Millington conducted a study [22] attempting to quantify the patterns of optic tract degeneration in subjects with occipital stroke. The study surveys 26 adult subjects and measures the percentage of damage to the occipital cortex and the corresponding laterality index. The relationship between the two is shown in Fig. 3.6.

Based on this relationship, I derived a linear equation linking the laterality index to the percentage of occipital cortex damage. I applied this equation to convert the known laterality indices from the Fahrenthold dataset into estimated lesion sizes. Given that the average surface area of human V1 cortical sheets is approximately 2643 mm^2 [23], I converted the percentage damage directly into lesion sizes (in mm^2) for each subject.

Before proceeding with the computational modelling, we assessed potential limitations in this lesion size estimation method. A fundamental expectation in stroke studies is that increased lesion size should correlate with reduced visual perception, measured as the subject's average accuracy in detecting stimuli across locations and luminance levels [37]. However,

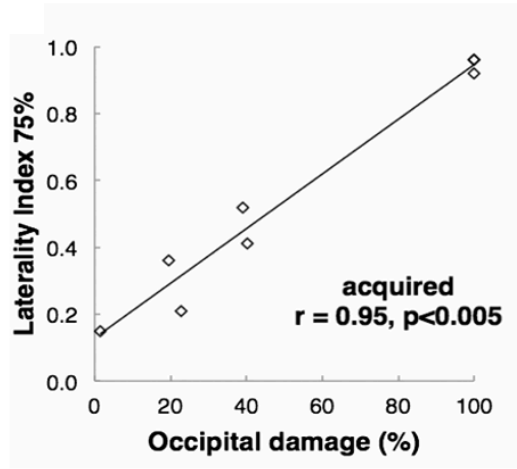


Figure 3.6: Graph showing the relationship between laterality index and percentage of damage to the occipital cortex. Figure adapted from [22]

plotting average subject accuracy against lesion size (Fig. 3.7) did not reveal the expected linear relationship. Instead, the plot indicates no significant linear correlation, as demonstrated by the low R^2 value.

This discrepancy suggests significant limitations in our current lesion estimation methodology. The primary reason for this may be that Millington et al. assessed lesions throughout the entire occipital cortex, whereas our dataset specifically involves lesions limited to V1. Therefore, the linear relationship identified between LI and occipital cortex damage percentage might not directly apply to our subjects. Nevertheless, these lesion size estimates serve as a valuable initial approximation. Future analyses should consider employing a broader range of lesion sizes to more accurately replicate individual patient behaviour.

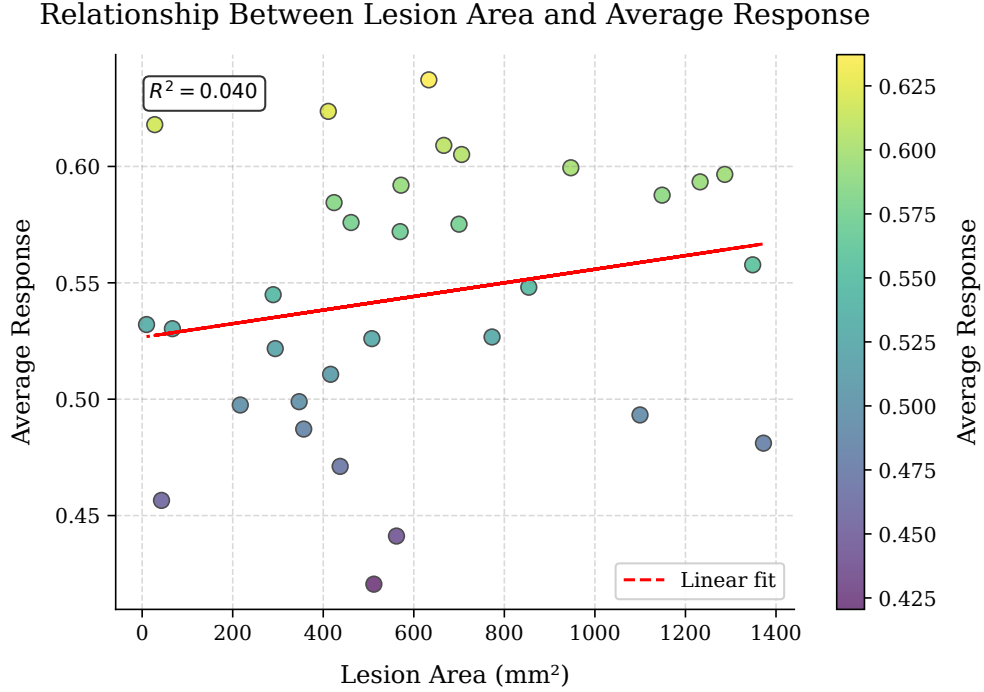


Figure 3.7: Average accuracy vs Lesion area

3.3. Modelling neural activity

A central assumption underlying this work is that layers within the TDANN can be meaningfully aligned with corresponding regions of the primate visual cortex. This is enforced by the network architecture, shown in Fig. 3.8. The TDANN is based on the ResNet-18 architecture [14], augmented by assigning each unit in the convolutional layers a position within a two-dimensional sheet representing cortical tissue. This spatial mapping enables a comparison with neural topography. The alignment between model layers and cortical areas is motivated by findings that earlier layers in deep convolutional networks tend to encode low-level features such as edges and textures, while deeper layers encode increasingly abstract visual information [15], which is reminiscent of the ventral visual stream. Following this, early TDANN layers are treated analogously to early visual areas like V1, while deeper layers correspond to higher-level regions like IT.

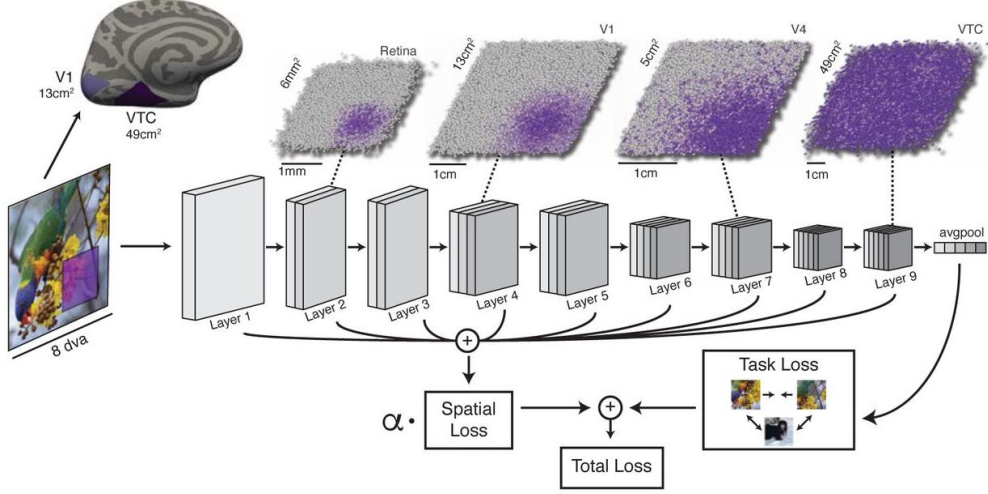


Figure 3.8: Architecture of the TDANN [4], which builds on ResNet-18 [14]. Units in each layer are mapped to a two-dimensional space representing cortical tissue. Figure adapted from [4].

The units are shuffled in the cortical tissue to ensure that nearby units have correlated responses. The positions of the units in the tissue are then frozen, and the network is trained with a self-supervised SimCLR task objective [16] and a spatial loss:

$$\mathcal{L}_{\text{TDANN}} = \mathcal{L}_{\text{task}} + \sum_{k \in \text{layers}} \alpha_k \mathcal{L}_{k, \text{spatial}} \quad (3.2)$$

where for the k^{th} layer, the spatial loss is given by:

$$\mathcal{L}_{k, \text{spatial}} = \frac{1}{M} \sum_{i=1}^M \left| r_i - \frac{1}{d_i + 1} \right| \quad (3.3)$$

Eq. 3.3 computes an average over pairs of units: r_i represents the Pearson correlation between the responses of a pair of units to a shared set of inputs, while d_i is the Euclidean distance between those units in the cortical sheet. Specifically, the objective encourages nearby units to have highly correlated responses while allowing uncorrelated units to be distant.

The TDANN learns spatially smooth representations by minimizing this objective, resulting in emergent functional organization reminiscent of cortical topography in the primate visual

system. This functional clustering is illustrated in Fig. 3.9. When trained without the spatial loss, this structure is degraded, and it is absent entirely in the unoptimized baseline.

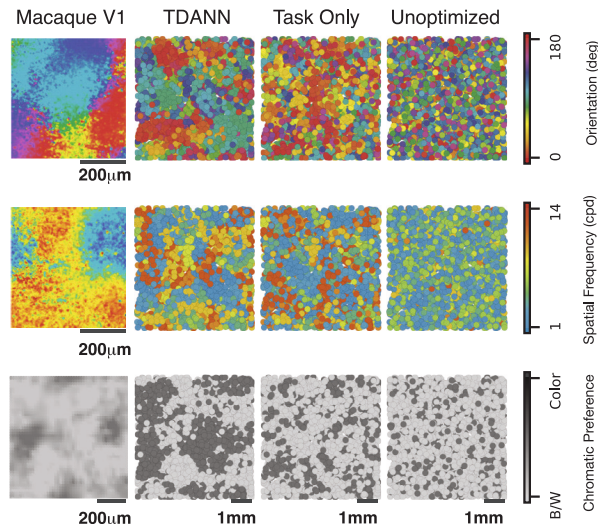


Figure 3.9: Comparison of the spatial organization of neural units in the model’s virtual V1 cortical sheet and macaque V1. The panels show that the topographic deep artificial neural network (TDANN) forms clustered representations for orientation, spatial frequency, and chromatic tuning, closely resembling the organization observed in the biological cortex. These clusters are markedly reduced in models trained without the spatial loss and are largely absent in the unoptimized baseline. Figure adapted from [4].

The functional and behavioural similarities of the TDANN to the visual cortex thus motivate our choice to use it for simulating strokes and lesions in the visual cortex.

3.3.1 Neural perturbations on TDANNs

A crucial criterion for the accuracy of ventral stream simulators is their ability to predict behavioural outcomes resulting from *perturbations of neural areas*. As highlighted by Schrimpf et al. [2], accurately modelling the behavioural consequences of such neural perturbations not only advances the development of more accurate biological models but also has direct implications for clinical applications, such as predicting and mitigating visual deficits in stroke patients.

TDANNs represent a significant advancement in this area. Specifically, Schrimpf et al. [2] demonstrated that targeted neural perturbations, such as electrical micro-stimulation, produce qualitatively similar effects in TDANNs and macaque IT cortex. Guided by this, our work aims to define perturbations in TDANNs that accurately mimic strokes in the visual cortex. By inducing these stroke-like perturbations in the model, I investigated whether the resulting behavioural changes align qualitatively and quantitatively with empirical observations from human stroke patients.

3.4. Interventions on TDANNs to simulate stroke

To simulate stroke-like perturbations in a TDANN, I first studied the characteristics of lesions formed by strokes in humans. The following figure illustrates a lesion formed by an ischemic stroke.

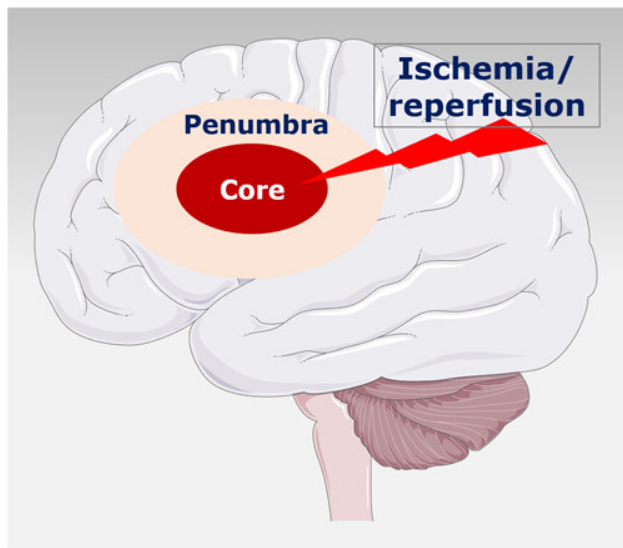


Figure 3.10: Ischemic Stroke, Figure adapted from [20].

Tuo et al. [20] found that the lesion volume in ischemic strokes is made up of two regions

- **Core:** This is a region that did not receive any blood flow during the blockage, and is completely made up of dead neurons

- **Penumbra:** This is a region that received limited blood flow and has damaged cells. However, it can theoretically be revived by reperfusion with blood.

The study found that if not treated immediately, which is rare, the core expands to encompass the full volume of the lesion.

Thus, for our modelling, we will assume that the entire lesion is made up of dead cells. To model this in the TDANN, we perturb the model layer mapped to the V1 area. This layer is divided into 70x70 arbitrary model units. Initially, we assume a one-to-one mapping between the model size and V1 size, i.e., 1 unit in the model corresponds to 1mm in length. Hence, to simulate a lesion of radius r mm, at a particular lesion location (x, y) , we zero out the coordinates in the model that are within a distance of r from the lesion centre (x, y) .

Some key assumptions being made here are outlined below:

- First, we assume that 1 unit of the model corresponds to 1 mm of distance in V1, which may not necessarily hold. Discussed in greater detail in [section 4](#)
- Second, we assume the lesions are circular, spanning the cortical sheets of V1.

3.5. Training the model

Now that we have perturbed the TDANN model to simulate the effects of a stroke lesion, we must test it on its ability to predict subject data. To do this, we use the activations from the last layer of the perturbed model, which correspond to the IT layer of the cortex, and train a linear classifier. The classifier is trained to provide a binary behavioural response to each stimulus, indicating whether or not it detected the stimulus. For further details on the training, refer to [Sec. 3.5.2](#).

3.5.1 Estimating behavioural data from Visual Field Maps

The visual field data provided by the Fahrenthold study is in the form of continuous visual field threshold maps. We must convert this to a discrete binary measure, an estimate of whether the subject would be unable to detect a stimulus at every given location and luminance level. We do this as follows:

Given an image with a stimulus at a particular location, and luminance (l) of a given subject, we navigate to the corresponding location on the visual field map, and obtain the threshold (t) at that location. Given that the visual field threshold is defined as the minimum luminance at which a subject detects the stimulus at least 50% of the time, and if the maximum luminance possible is $l_{\max}$, we can quantify a probability p that the subject detects the stimulus as:

$$p = \begin{cases} 0.5 + \frac{l - t}{l_{\max} - t}, & \text{if } l > t \\ \frac{l}{t}, & \text{if } l < t \end{cases}$$

We sample from a Bernoulli distribution with parameter p to obtain the subject's behavioural response.

3.5.2 Linear classifier

To evaluate the effects of simulated stroke lesions on visual perception, I generated images matching the 10-2 and 24-2 grid patterns used in HVF tests, alongside null (blank) images. These images serve as inputs to the perturbed model, whose IT cortical activations are subsequently processed through a trainable linear classifier.

This linear classifier, implemented as a single fully connected layer, is trained to differentiate between images containing visual stimuli (assigned labels of 1) and null images (assigned labels of 0). The rationale behind this approach is that an effective stroke lesion simulation should significantly impair neuronal units in the model. Consequently, the linear classifier's

accuracy in detecting stimuli at specific visual field locations should diminish, reflecting the compromised neuronal representations.

The classifier is trained using binary cross-entropy loss and optimized by the Adam optimizer [21] with a learning rate of $1e-2$. The training procedure is executed independently for each lesion location and subject pair. To ensure reliable estimates of accuracy, we employ 5-fold cross-validation, partitioning the dataset based on stimulus locations.

It is to be noted that a separate linear classifier is trained for each subject, and lesion locations, for both 10x2 and 24x2 grid types. As mentioned earlier, the TDANN has a 70x70 cortical sheet which corresponds to V1 in the brain. Thus, we explore lesion locations with both x and y coordinates ranging from 5 to 65 units, in increments of 5.

Chapter 4

Results

As described in Sec. 3.4, we simulate a stroke-like lesion in the V1 layer of the TDANN and assess the model for qualitative and quantitative parallels with human stroke patients. Behavioural responses are generated by applying a linear readout to the IT-layer features, yielding a binary decision: stimulus detected or not detected. If the model’s accuracy at a specific stimulus location falls, it indicates that the IT features provide less useful information for stimulus detection at that location, analogous to a stroke patient failing to perceive the stimulus.

It is important to note that while lesion sizes were estimated for the human subjects, the corresponding area these occupy on the TDANN’s cortical sheet is not precisely known and must be approximated. To account for this uncertainty, I explore a range of scaling factors when mapping lesion areas to the model. The specific scaling factor used is noted in all relevant analyses where ambiguity may arise. For reference, I assume that each unit in the TDANN’s 70×70 cortical sheet corresponds to 1 mm^2 of cortical area. A scale factor of x therefore increases the perturbed lesion area in the cortical sheet by a factor of x in surface area.

4.1. Model accuracies

I began by training models on behavioural data from an initial subset of five subjects, covering all lesion locations described in Sec. 3.5.2. This subset was chosen due to the computational cost of storing activations across all subjects simultaneously. By first analyzing these five subjects, I aimed to evaluate the impact of stroke-like perturbations and identify meaningful patterns in the model’s responses before scaling the analysis to the full dataset.

For each subject and lesion location, I computed the model’s detection accuracy across a range of lesion scale factors. These accuracy values were then interpolated across lesion locations to produce the heatmaps shown in Fig. 4.1.

First, we can observe that model accuracy remains relatively high across most subjects, even as the lesion size increases. Notably, only for subjects CB10 and CB11 does the accuracy drop substantially: approaching chance-level performance (approximately 50%) when lesions are centered in specific cortical regions. This is consistent with expectations: if the simulated lesions are functioning as intended, we should observe chance-level accuracy in at least some areas of the visual field for sufficiently large or well-placed lesions.

Second, as previously discussed, neurons in the visual cortex are spatially organized such that functionally similar units are clustered together [4, 8]. Stroke-induced damage to localized cortical areas is therefore expected to impair groups of neurons with shared functional properties, leading to continuous, spatially localized deficits rather than randomly distributed impairments.

And as expected, the heatmaps do indeed show smooth, spatially coherent variations in accuracy as lesion location and size vary. This smoothness demonstrates that functionally related units are indeed spatially clustered within the model cortex, closely mirroring known properties of biological visual cortex organization. Darker patches in the accuracy maps reveal critical cortical areas, indicating regions whose neuronal units are especially important for detecting the given stimulus.

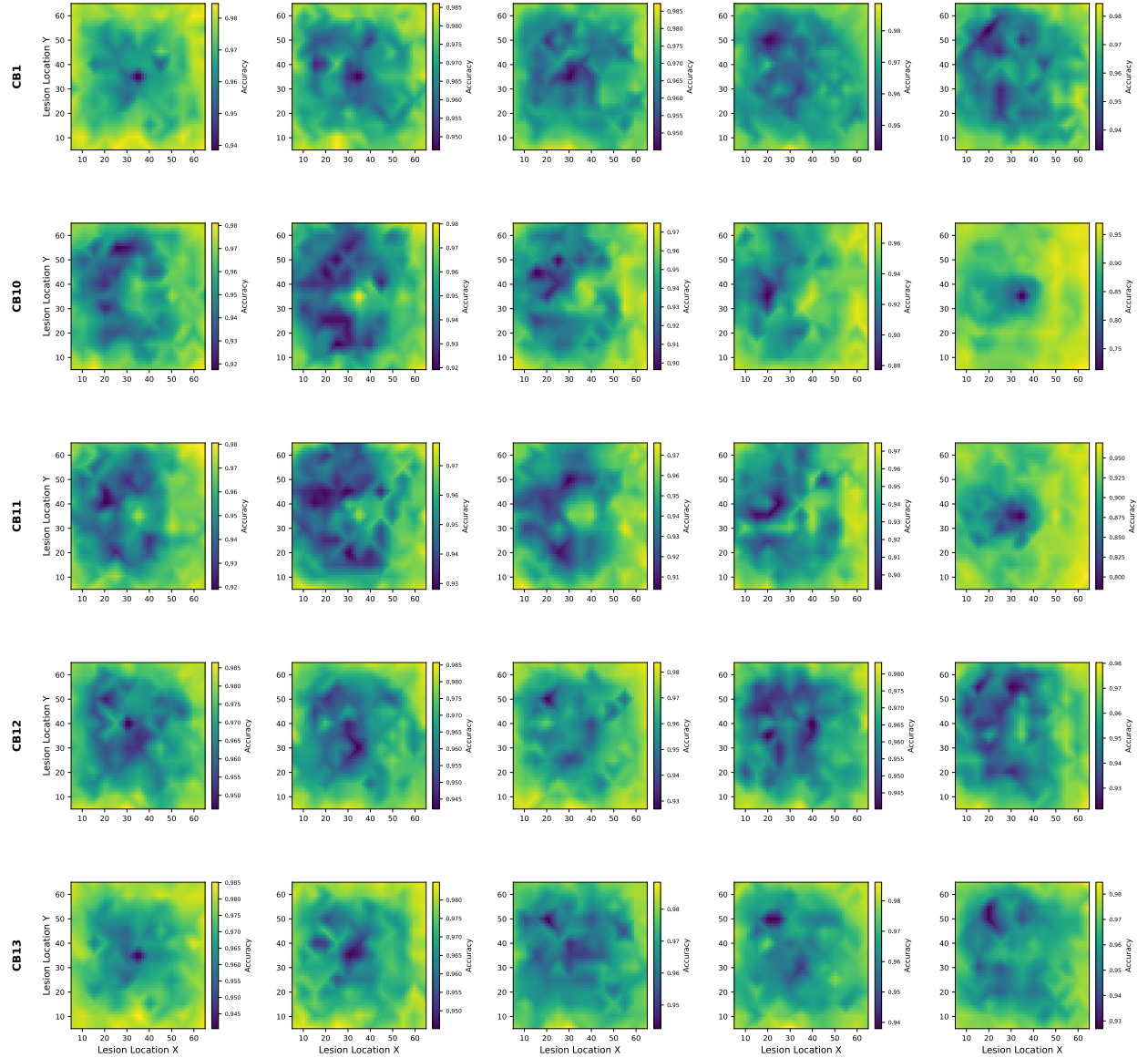


Figure 4.1: Average detection accuracy of the TDANN after simulated lesions (stroke-like perturbations) of varying sizes centered at different locations in the V1 cortical sheet. Each map corresponds to a different lesion scale factor, with darker regions indicating greater performance drop. The smoothness of these maps suggests that functionally similar units cluster spatially, as lesions centered in certain areas cause consistent accuracy degradation. The maps depict results from five subjects, with lesion scale factors increasing from left to right: 1.0, 1.25, 1.5, 1.75 and 3.

4.2. Qualitative analysis

4.2.1 Effect of the stroke perturbation

From the previous plots, we already have the model’s accuracy for each subject, lesion location, and scale factor. To assess the worst-case perceptual impact, I focus on the *minimum* accuracy observed across all lesion locations for each subject and scale factor. As seen earlier, the average accuracy remains high; typically around 90% across most lesion sites. Therefore, the minimum accuracy provides a more informative measure of how severely the model’s performance can be disrupted by a lesion in the most vulnerable region of the cortical sheet.

Now, we know biologically that increased lesion areas in V1 lead to more severe visual perceptual impairments. [37]. As shown in Fig. 4.2, there is a clear trend in the TDANN: enlarging the radius of the simulated lesion leads to a reduction in average detection accuracy, mirroring the pattern in the brain.

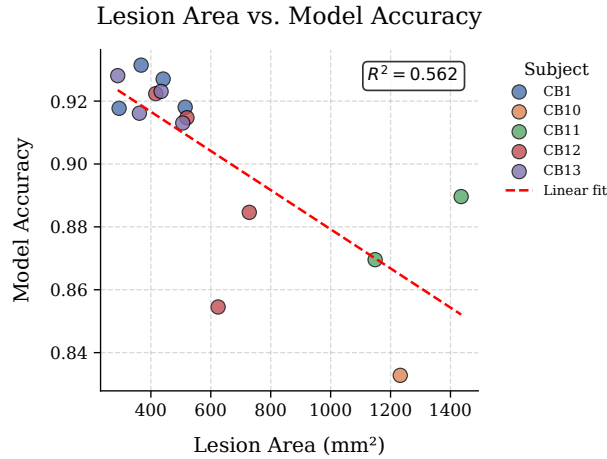


Figure 4.2: Model accuracy under varying lesion sizes. For each of the 5 subjects and multiple lesion scale factors, I simulate lesions across all possible cortical locations and report the *lowest* resulting accuracy for each lesion size. This reflects the worst-case perceptual impact of lesions of increasing severity.

4.2.2 Anatomical accuracy

In addition to accurately simulating behavioural outcomes, an essential component of a realistic visual cortex model is anatomical consistency, particularly in how it replicates the effects of a stroke.

I examined this anatomical correspondence by systematically placing stroke-like lesions at nine distinct cortical positions: top-left, top-middle, top-right, center-left, center, center-right, bottom-left, bottom-middle, and bottom-right (see Fig. 4.3). The lesion size was chosen to be that of CB10, with a 3x scaling factor, because its global accuracy values indicated that the lesion was large enough to be having an effect on perception. For each lesion location, I assessed the model’s accuracy across stimulus luminances at all visual field locations. The visual cortex is known to be retinotopic, meaning cortical neuron locations correspond systematically to visual field positions [17]. Thus, in a biologically accurate model, we expect distinct lesion locations to selectively impair detection at corresponding regions in visual space; for instance, a lesion at the top-left cortical region can primarily reduce accuracy in detecting top-left stimuli.

However, contrary to expectations, the TDANN model did not display clear retinotopic specificity following stroke perturbations. Notably, lesions near the center of the cortical map disproportionately reduced overall accuracy, whereas peripheral lesions had significantly weaker effects. Although the design in [4] aims explicitly to preserve retinotopic mappings by assigning neuronal units based on their spatial receptive fields, I hypothesize that the subsequent unit shuffling and spatial loss optimization process disrupts this intended structure.

To summarize, while TDANN captures some key aspects of anatomical realism, such as local functional clustering consistent with cortical organization, it falls short in preserving precise retinotopy. This discrepancy highlights an important consideration for future developments and refinements of these models.

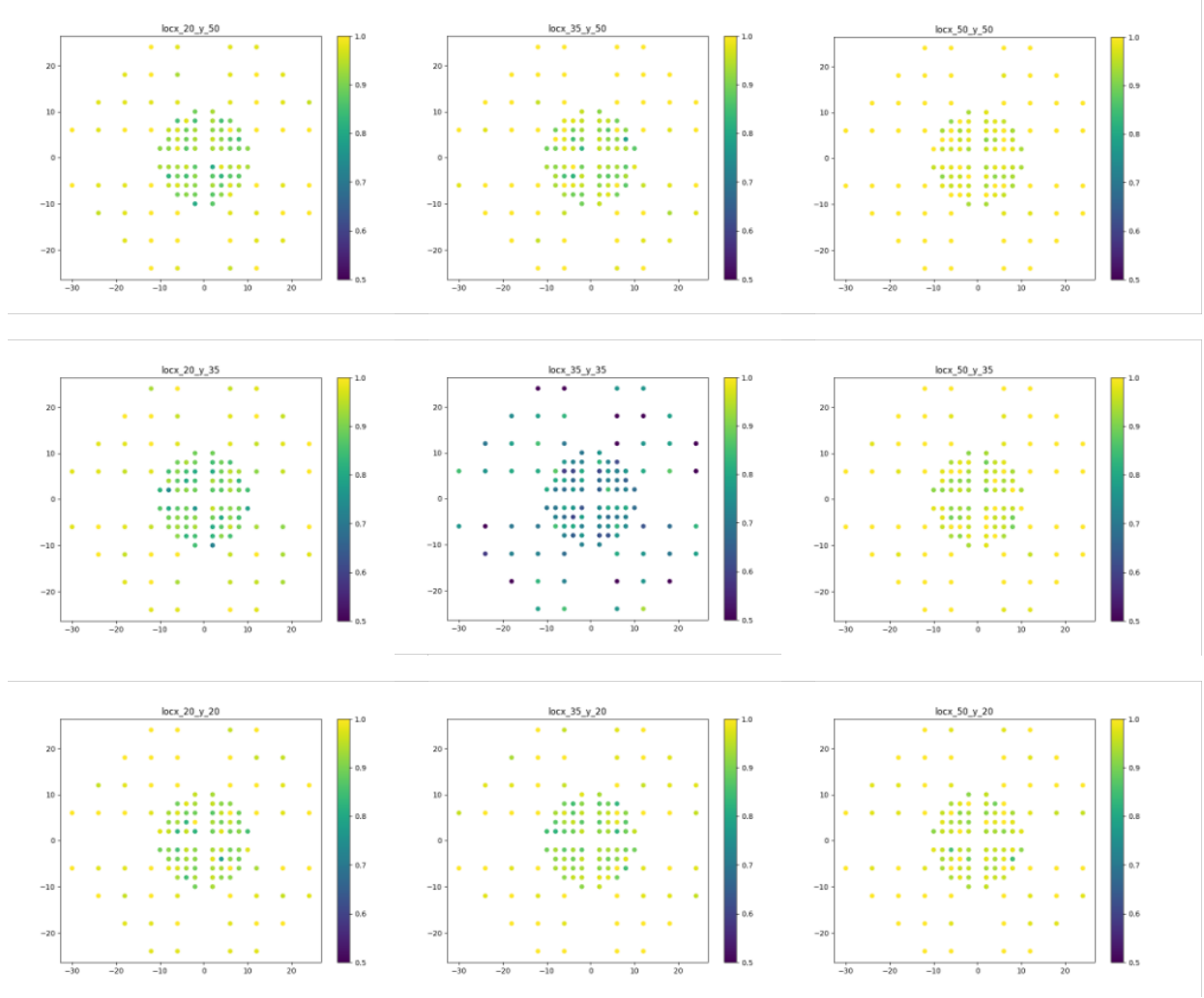


Figure 4.3: For each of nine lesion epicenters arranged across the cortical sheet (top-left, center, bottom-right, etc.), I plot the model's accuracy across all stimulus locations. In a retinotopically organized system, different lesion positions should impair detection of different spatial stimuli. However, the model predominantly shows a performance drop when lesions are centered near the middle of the cortical sheet, suggesting a lack of precise retinotopy. A lesion scale factor of 3.0 was used for the plots.

4.3. Quantitative analysis

Despite qualitative agreements, paralleling [2], the results indicate significant discrepancies in the quantitative predictions of the model. Specifically, while the accuracy for human subjects in detecting stimuli ranged from 42.05% to 63.72%, the TDANN required substantially larger lesion scale factors (in some cases, scale factors of 5 or even 10) to achieve similarly impaired performance levels, such as the trivial 50% detection accuracy in some locations. For smaller-scale factors, the IT embeddings still contain sufficient information to support an accurate linear readout. A plausible explanation for this quantitative mismatch could be neuronal redundancy within the TDANN: even when lesions disable functionally clustered neurons, unaffected neurons still retain enough information to allow for reliable stimulus detection.

Chapter 5

Discussion

In this thesis, I analysed the potential of a biologically inspired convolutional neural network, the TDANN, to simulate stroke-induced visual impairments in the primate visual cortex. Through neural perturbations in the model that mimic lesions observed in stroke patients, I assessed the qualitative and quantitative agreement between model behaviour and the empirical data from human subjects.

5.1. Summary of results

Our experiments demonstrate that the TDANN successfully replicates certain qualitative features observed in stroke-induced visual impairments. Specifically, I found that increasing the simulated lesion size in the TDANN resulted in a consistent decline in visual detection accuracy, mirroring the relationship seen in stroke patients. The analysis also showcased that functionally similar neuronal units within the TDANN cluster spatially, as evidenced by the smoothness of the accuracy maps obtained from various lesion locations. I stress that the qualitative alignment is obtained **without model access to the stroke data**: this alignment therefore showcases the accuracy of the TDANN to simulate the ventral stream.

However, despite qualitative successes, there are clear quantitative discrepancies. The

TDANN requires substantially larger lesion scale factors than estimated human lesion sizes to reach comparable reductions in detection accuracy. This misalignment points toward significant redundancy in the model’s architecture; even extensive localized perturbations leave sufficient residual neuronal activity for accurate stimulus detection.

5.2. Contributions and limitations of this study

This thesis contributes to the field of computational neuroscience and stroke rehabilitation by demonstrating the promise of goal-driven TDANNs to accurately simulate the primate visual cortex, behaviourally and anatomically. The qualitative replication of various stroke-induced behaviours provides a proof-of-concept for using these neural nets as biologically plausible models of cortical lesions. The thesis also highlights *specific directions for improving the modelling*. For instance, future architectures should focus more on maintaining retinopathic mappings for better qualitative agreement, and perhaps a more aggressive functional clustering for better quantitative agreement.

Despite the strengths, there are several important limitations that I highlight here:

- **Quantitative discrepancy:** The inability of lesions to more significantly impact model accuracy indicates a lack of precise one-to-one mapping between the TDANN’s virtual cortical sheet and actual human cortical organization.
- **Loss of retinotopic organization:** As highlighted earlier, despite architectural design intended to maintain retinotopic mappings, we observed a loss of precise retinotopy, possibly due to initial shuffling and the subsequent optimization procedure.
- **Simplified lesion modelling:** Not all lesions are created equal, as noted in Sec. 3.4. The modelling in this thesis is simplistic, and the neuronal deactivations are solely a function of lesion size. Nonetheless, this work represents a novel application of TDANNs to the study of stroke and serves as an initial, approximate method. It provides a foundational step toward more sophisticated modelling in future research.

- **Lack of patient-specific lesion modelling:** Another major issue with our modelling is the lack of access to patient specific data on the lesions, both in terms of lesion sizes and locations. This poses a major challenge, especially in our ability to predict subject specific behavioural data.

5.3. Future directions and conclusion

In addition to the directions outlined in the previous sections, I describe other promising avenues:

- **Modelling complex lesion patterns:** Extending the lesion perturbation to allow for irregularly shaped or multi-region lesions would provide greater realism. This encompasses both simulating multiple lesions in the same layer, as well as across layers in the model.
- **Linear readout from earlier activations:** The current behavioural readout relies on the IT layer, but much of the visual processing occurs in earlier cortical areas. Investigating linear readouts from pre-IT activations (for instance, V4 or V2) could yield further insights.
- **Localizing lesion locations:** This framework could be adapted to localize lesions within the TDANN’s cortical sheet. By performing a grid search over potential lesion sites and correlating predicted impairments with observed subject data, it may be possible to estimate lesion locations precisely; particularly once the modelling is more accurate and tuned to preserve retinotopic structure.
- **Subject specific modelling:** A comprehensive gridsearch could be conducted for each subject, spanning a range of lesion locations as we have already done, but also a range of lesion sizes, apart from the baseline estimates.
- **Temporal realism:** Stroke outcomes vary significantly across different recovery phases [1]. Incorporating temporal dynamics, such as differences between subacute and chronic

stages, could enhance biological realism.

In conclusion, this thesis illustrates the potential of Topographic Deep Artificial Neural Networks as tools for understanding and simulating cortical impairments due to strokes. While the experiments highlight a need for continued refinement, by bridging computational neuroscience and clinical neurology, this line of research enriches our understanding of stroke-induced visual deficits and lays the groundwork for computationally informed clinical interventions.

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